Power BI

The Corporate Training by Abhimonyu Dey

Course Outline

- **Day 1: Introduction and Overview of Power BI**
- **Day 2: Data Modeling & Relationships**
- **Day 3: Data Visualization & Report Design**
- **Day 4: Advanced DAX & Time Intelligence**
- **Day 5: Publishing, Sharing & Power BI Service**

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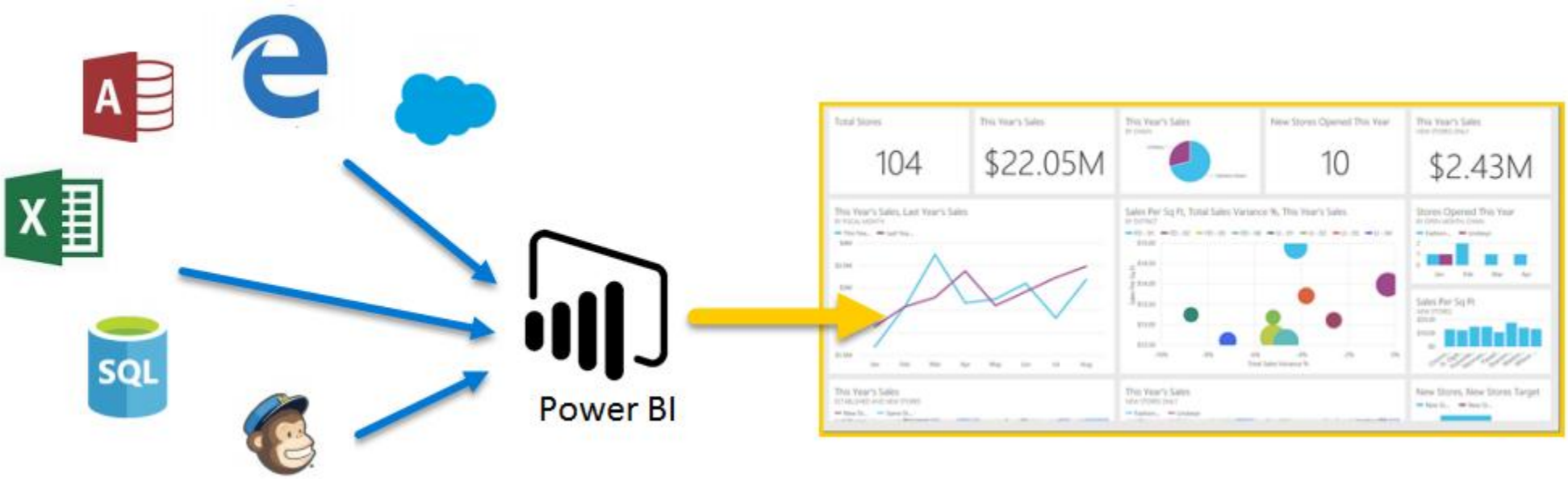
What is Power BI

- Power BI is a robust business intelligence tool developed by Microsoft that allows users to transform raw data into meaningful insights through interactive visualizations and reports. It's made to be easy to use, so people with different levels of technical skill are able to create and share data-driven insights. Power BI can connect to many types of data sources, such as Excel files, cloud storage, and local databases, making it a flexible tool for analyzing data.



Benefits of Power BI

- **Integration with Microsoft Products:** Power BI works proficiently with other Microsoft tools like Excel, Azure, and SharePoint, making it easy to use with your current IT systems.
- **User-Friendly Interface:** Its simple design helps everyone, no matter their skill level, to easily make and share reports and dashboards. The drag-and-drop feature and simple language questions make it easy for newbies to use.
- **Real-Time Data Processing:** It lets you view and analyze data in real-time so that your decisions are based on the latest information and up-to-the-minute.
- **Advanced Data Visualization:** The platform offers exceptional data visualization capabilities that let users create interactive and visually appealing reports and dashboards in just a few minutes.
- **Data Security:** Power BI has strong security features. It includes end-to-end encryption and row-level security and follows data protection rules to keep data safe for all users and devices.
- **Cost-Effective:** Power BI has different price plans, including a free version, which makes it available for businesses of any size.
- **AI and Machine Learning:** Power BI includes built-in AI features like AutoML and AI Insights, which allow users to create machine learning models and gain insights from structured and unstructured data.
- **Integration with Diverse Data Sources:** It connects to many different data sources, which include databases, cloud services, and live streaming data, to provide a complete picture of how well the business is doing.
- **Frequent Updates:** Microsoft updates Power BI frequently to add new features and make improvements so that users can always access the latest tools in data analytics.
- **Collaboration and Sharing:** Power BI allows teams to work together in real time and share findings, making teamwork more effective.

Power BI use cases

1. Resource Management: Power BI, with strong visualization capabilities, can be used to make better decisions related to task and resource allocation. Businesses use “Power Pivot” and “Power Query” to generate reports. With that, you can track the major KPIs like task count, number of open projects, project completed, and resource skills. The tool is applicable to the manufacturing sector, which has multiple production facilities in numerous locations.

2. Sales Scorecard: Managing salespersons is as hard as managing leads. Understanding if each individual is doing well determines the success of the sales initiatives. With Power BI, individual sales reports can be developed and viewed to track team performance. Slide and dice reports are reviewed based on the products, team performance, and geography-wise located salespeople.

3. Financial Reporting and Analytics: With the mindful use of the data visualization tool Microsoft Power BI, businesses have streamlined financial tracking. A financial reporting dashboard is often used to understand complex financial figures by having a bird’s-eye view of revenue streams and expenses. This would assist in quick decisions pertaining to cash flow through a product-wide data summary.

4. Claims, Billing, and Collection Report: Power BI tool will play a central role in the claim management process. It develops customized dashboards and data summaries that can help analyze claim data. The real-time view of the reports allows the claim processing team to communicate with the team and evaluate results. The insights derived from the data analysis can create a seamless experience for both customers and employees.

5. Supply Chain and Inventory Management: Microsoft Power BI can be integrated with supply chain operations to leverage the monitoring and optimization of inventory levels. It provides sophisticated dashboards for visualizing inventory levels and does demand and supply analysis that boosts decision-making. The real-time insights help operational teams identify bottlenecks and monitor supplier performance to maintain efficient supply chains.

6. Customer Segmentation: Even if you have an idea of who your customers are, you still lack the information for product positioning. When you use Power BI, you can attain a deep understanding of the customer segment by analyzing the behavioral patterns in specific groups. The customer segmentation dashboard allows you to prioritize the new products while focusing on the customer groups who are interested in your offering.

7. Project Performance Tracking: Power BI has various project management tools like Trello and Microsoft Project, which allow project monitoring by tracking key milestones. Managers can get real-time information about the project status, team performance, and resource utilization. They can identify potential risks by assessing the centralized dashboard that contains metrics for project progress. It leads to improved decision-making and better alignment of budget across multiple projects.

8. Customer Service Analytics: Quality customer service is regarded as a key to customer satisfaction, which is why organizations prioritize it. Power BI contains customer service analytics dashboards from which businesses can leverage several aspects of customer service. Metrics like customer feedback scores, customer satisfaction (CSAT) scores, sentiment analysis, and issue resolution rates are regularly measured to address service gaps promptly.

9. Student Performance Tracking: Educational institutes utilize the Microsoft Power BI tool to analyze the performance of students and determine enrollment trends. It is not only used for academic performance, but it is also used for tracking the attendance and learning curves of students to examine the level of attention given to students to gain excellence. Power BI will help teachers recognize students with low productivity so that interventions can be established for their academic success.

Who Can Use Power BI?

1. Business Analyst: Business analysts focus on deriving solutions for business problems by monitoring business operations. Data intelligence tools like Power BI can be a good companion for their analysis and interpretation. With power data modeling and a drag-and-drop interface, Power BI can help business analysts determine relationships between datasets so that they can make recommendations accurately.

2. Data Analyst: Data analysts routinely play with data. They perform statistical analysis through predictive modeling of data to convert it to layman's terms. Power BI can be a helpful tool for visualizing data and deriving actionable insights. This tool can be integrated with other machine learning models to develop an intuitive dashboard that helps data engineers with their dynamic reporting needs.

3. Marketing and Sales Analyst: Marketing professionals are indulged in increasing the number of leads through running campaigns, tracking sales metrics, and planning on customer acquisition. Power BI, when integrated with other tools like Google Analytics and CRM systems, allows marketers to track daily metrics like a pro. They can understand campaign performance and measure customer engagement.

4. Operational Managers: Operational managers focus on managing routine business operations and suggesting solutions for prudent resource utilization and cost reduction. With active visualization offered by Power BI, operational managers can stream data and obtain real-time insights that help them stabilize supply chain performance. Actionable insights enable optimized processes and prompt maintenance that improve productivity.

5. Project Managers: Project managers work day and night to ensure the successful delivery of projects. For that, they plan and analyze, distribute tasks, allocate resources, and optimize budgets. With the help of Power BI, they can assess the project's health by tracking down the key metrics. The dashboards will give a centralized view of all deliverables, which they can track and establish interventions if any deviations are found in the process.

6. Management-Level Executives: People at the management level are involved in making strategic decisions to bolster the financial performance of the enterprise. With Power BI as their companion, they can leverage their decision-making, which is backed by data. Power BI dashboard contains a comprehensive analysis of business KPIs that allows managers to undertake data-driven decisions. This means they would be able to exercise responsive leadership to make a strict alignment of decisions with the goals.

Companies That Use Power BI

1. MediaCom: MediaCom is the largest cable TV provider that needs to measure diversified datasets across multiple media channels. In the urgency of increasing the feasibility of analyzing data from different sources, Power BI provides single-sourced data gathered from numerous sources. Customized dashboards of Power BI enabled them to analyze different metrics such as audience demographics and ROI. Eventually, the company spent less time analyzing individual reports as they had a comprehensive single report that dictated the impact of campaigns and ad performance.

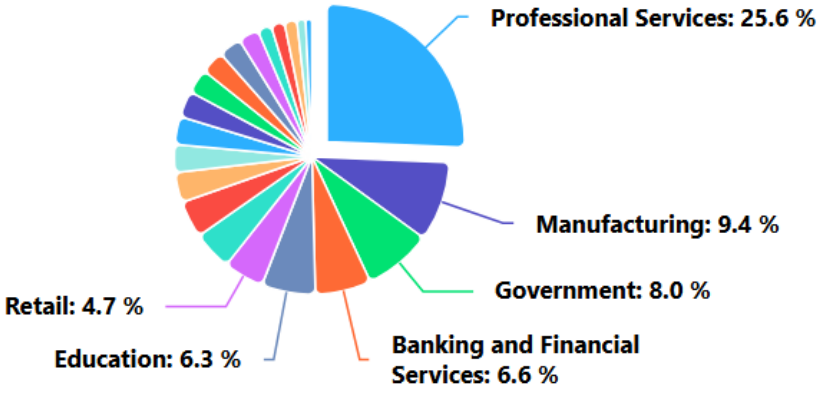
2. Heathrow: Heathrow Airport UK has adopted Power BI to streamline management and airport operations. The tool helps visualize data from numerous areas, like checkpoints and luggage handling, to obtain real-time information regarding traffic in the passenger flow. This gives employees an idea of how to proactively manage traffic conditions to facilitate a less stressful traveling experience for passengers. With the data intelligence, the company runs Airport operations more smoothly than ever before.

3. Walmart: A giant retail chain, Walmart, attempts to use Power BI to manage operational data across its vast network of stores. The company has complex supply chains due to the many stores, which creates difficulty not only in gathering data but also in measuring performance. With Power BI, they transformed their data management approach by bringing a single analysis platform for a unified view. By getting access to real-time issues, the company strategized for a risk management plan that dealt with issues promptly. In terms of its outcomes, predictive analytics made inventory management possible and empowered employees to undertake data-driven decisions.

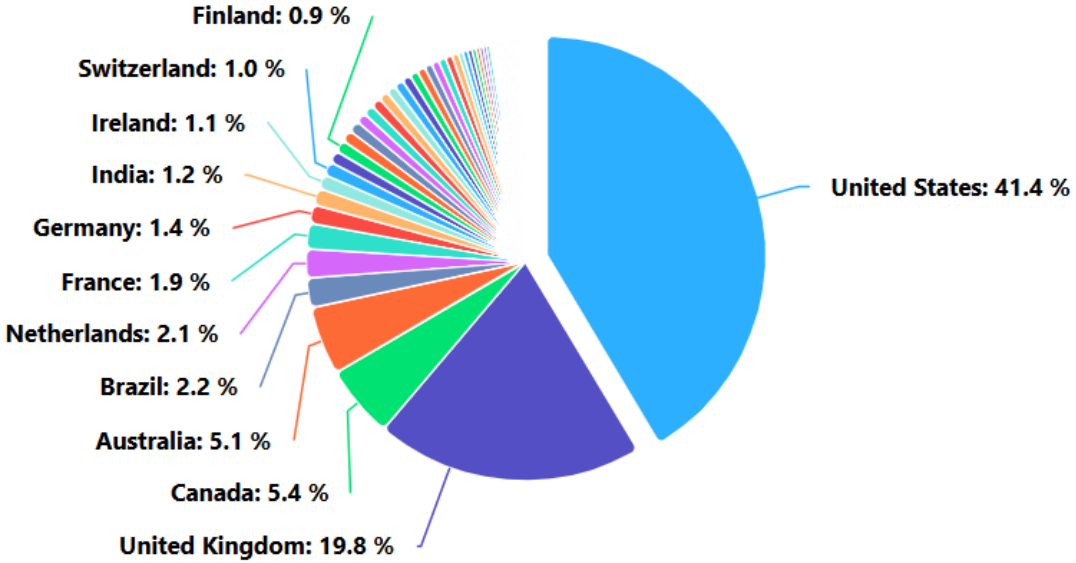
4. Metro Bank: The challenge with Metro Bank was not being able to track its financial performance and establish standards for reporting compliances. With the help of the Power BI tool, the company created financial dashboards for tracking down real-time data and also measuring the financial metrics. As a result, the reporting time for Metro Bank was reduced by 30%. The accurate reporting and streamlined data management allowed fewer errors. Employees also got freed up from managing individual reports and emphasized more on strategic planning.

Customers of Power BI

Breakdown of Microsoft Power BI customers by Industry

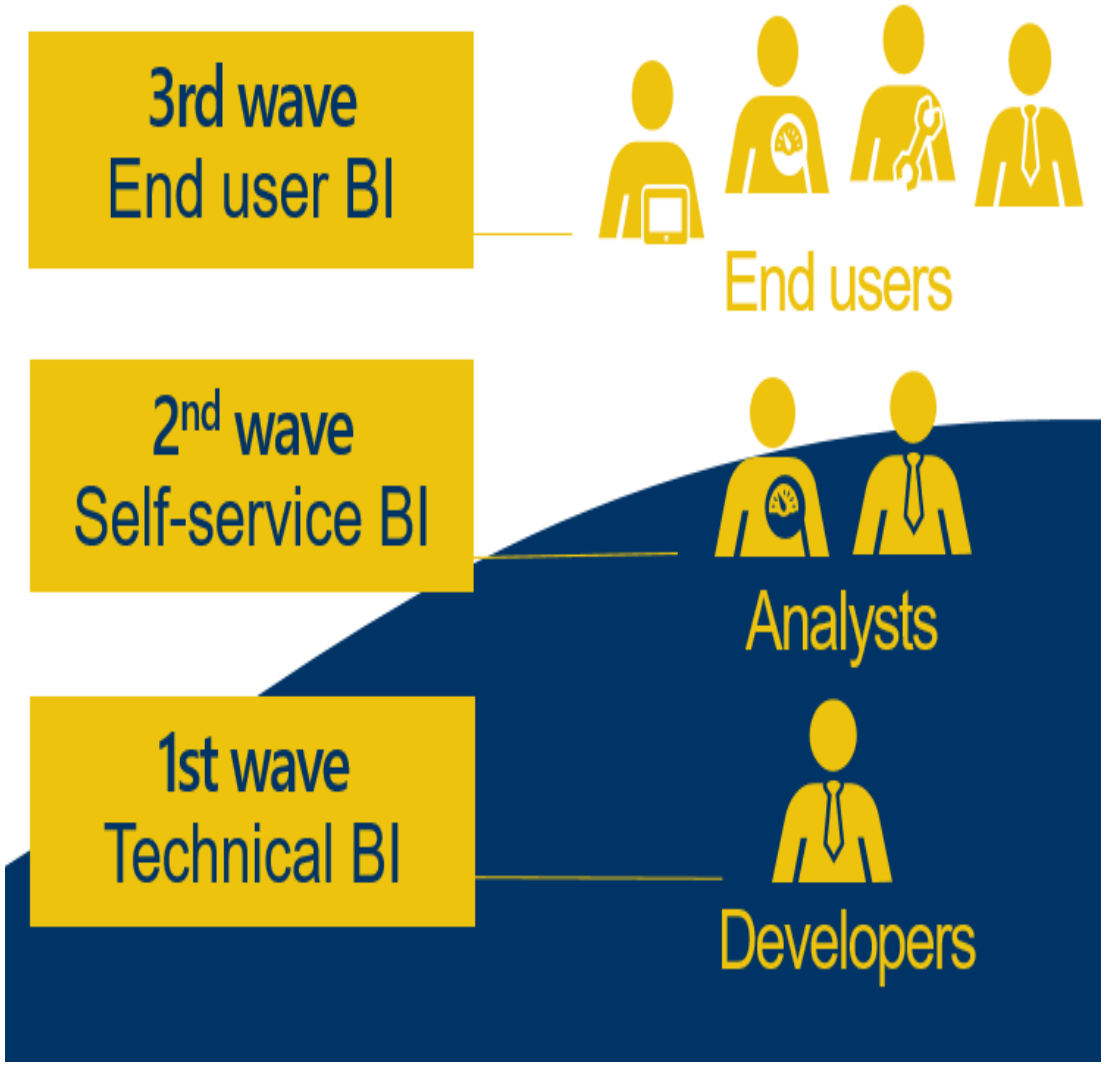


Breakdown of Microsoft Power BI customers by Country



Power BI Roles and Users

- **Developer:** For the developers, there is a need to have an understanding of basic database relations. It is important to know why too many relations cause bad data and how primary keys function. The developers also need to have an understanding of how to pull data from where it is stored and then how to transform and clean it into usable data. PowerBI has low to no code wizards to help these transformations, but an understanding of what needs to be done is very important. This role will also be responsible for creating all the DAX calculations and measures, so PowerBI developer license users will need knowledge of the DAX language. The developer will also need to be creative because one of the benefits of PowerBI that these reports start on a blank canvas.
- **Analyst Users:** The analyst role needs to have a basic understanding of the data from a technical standpoint. They should also know how tables relate to one another so that they can build meaningful visuals. Unlike the developer, they only need basic technical knowledge and will not need to make or change any of the relationships or data manipulations as those are done by the developer. The analyst will also need to have a bit of a creative side. For the most part, they will only be editing reports that are provided to them, but there are often times that one-off and ad-hoc reports are needed and this is a job for the analyst. They can use the data-sets already shared with them to build these reports, which not only saves time, but they also have the reassurance that the data they are using is correct and up-to-date. All they need to do is drag and drop the field that they want to use in their new visuals from the powerbi.com designer interface and then share the reports they have built. No coding or SQL knowledge needed.
- **End User:** The last of the three PowerBI user types is the end user. The end user should know how to navigate the powerbi.com web application interface. This includes how to filter and slice data and navigate from report to report. Not much technical knowledge is needed for the end user role.



Power BI update timeline

- **2011.** Power BI was originally called Project Crescent and was initially available in July 2011, bundled with Microsoft SQL Server, code name Denali.
- **2013.** It was renamed and unveiled in September 2013 as Power BI for Office 365 -- now Microsoft 365.
- **2015.** Initially based off Excel features like Power Query, Power Pivot and Power View, Microsoft added features over time, including enterprise-level data connectivity and security options. Power BI was released as a standalone product in July 2015.
- **2017.** The Microsoft Power BI desktop app was released in the Windows store.
- **2019.** Power AI released AI-based enhancements and previewed Key Influencers, its first AI-powered visualization feature.
- **2020.** Power BI Desktop users could access AI insights through Power Query.
- **2023.** Microsoft announced generative AI features for Power BI through integration with Microsoft's Copilot.

Power BI Licenses

Power BI License	Cost	Storage	Dataset Size	Data Refreshes	Collaboration	AI/Advanced Features	Report Sharing	Who It's Best Suited For
Power BI Desktop (Free)	Free	1GB cloud storage	1GB	N/A	No	R and Python integration	Cannot share via the cloud	Individual users who don't need to share or collaborate
Power BI Pro	$10 per user per month	10GB per user	1GB	8 scheduled refreshes per day	Share reports with other Pro users	No advanced AI features	Share reports within Power BI service	Small teams that need collaboration and sharing capabilities
Power BI Premium	Per user: $20 per user per month	100GB per user	100GB	Per user: 48 refreshes per day	Share with free users and external stakeholders	Advanced AI and machine learning capabilities	Share with external stakeholders without requiring licenses	Organizations needing advanced features, analytics, and large-scale sharing
Microsoft Fabric SKU	Varies by capacity (replaces Premium Per Capacity)	Scalable (Azure-based)	Scalable datasets	Real-time analytics	Share across organizations, integrates with Azure	Advanced AI, multi-workload support, and Azure-only features	Seamless sharing within Azure environment	Enterprises needing unified data analytics and real-time collaboration

Comparative Analysis: Power BI vs. Other BI Tools:

1. Tableau

- **Strengths:**

1. It is renowned for its robust data visualization capabilities.
2. Offers a wide range of pre-built connectors for various data sources.
3. Strong community support and a plethora of online resources.

- **Considerations:**

1. Higher learning curve compared to Power BI.
2. Licensing costs can be higher, especially for larger deployments.

2. QlikView/Qlik Sense

- **Strengths:**

1. The associative data model allows for dynamic and exploratory analysis.
2. Robust in-memory processing for faster data retrieval.
3. Strong visualization and dashboard creation capabilities.

- **Considerations:**

1. Requires substantial development effort for complex applications.
2. Licensing costs can be on the higher side.

3. Looker

- **Strengths:**

1. A cloud-native BI tool with a focus on data exploration and collaboration.
2. Utilizes LookML, a data modeling language, for centralized and version-controlled data definitions.
3. Suitable for organizations with extensive cloud data sources.

- **Considerations:**

1. Learning LookML may pose a challenge for some users.
2. The pricing structure can be complex.

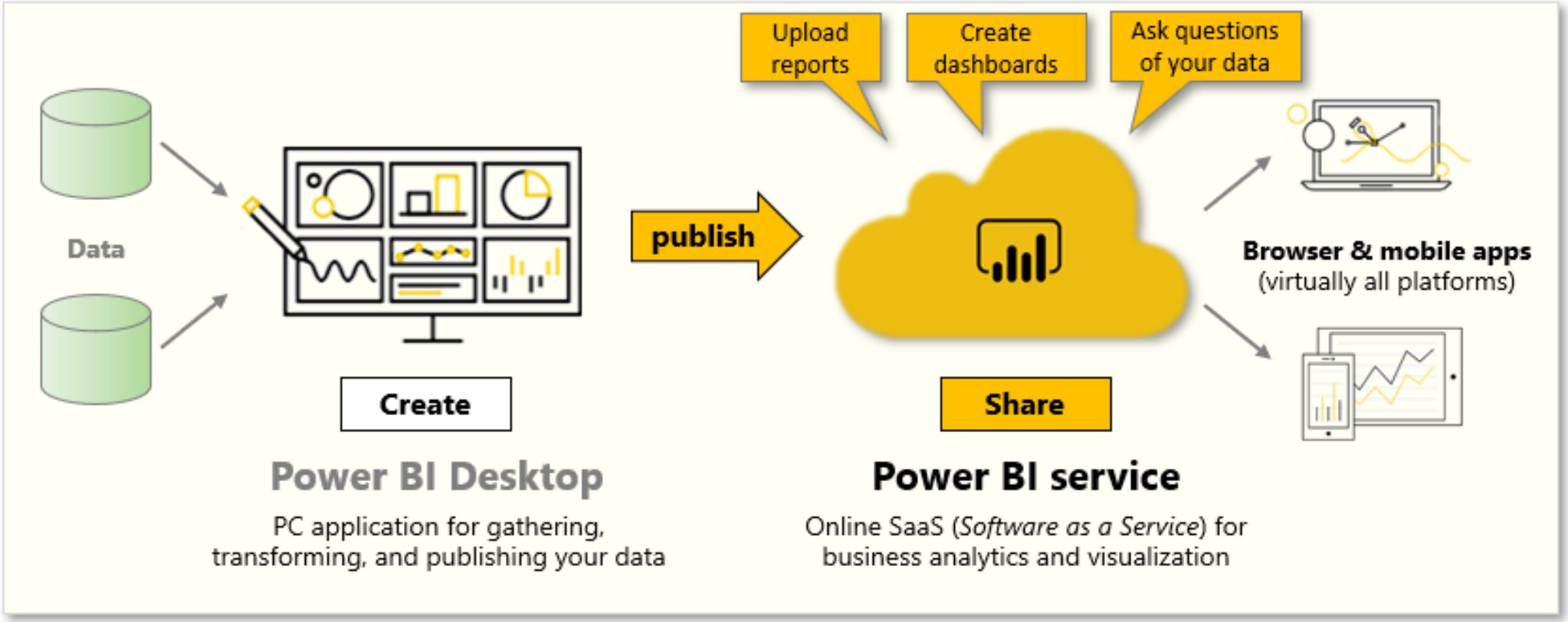
Power BI vs Excel

Feature	Power BI	Excel
Strengths		
Calculation Speed	Fast, handles large datasets	Fast, easy for basic and complex calculations
Use Cases	Data analysis, visualizations, KPIs, dashboards	Versatile (accounting, HR, operations, business analysis)
Visualization	Interactive, drop-and-drag, wide variety	Limited visual options, mainly tables and charts
Collaboration	Real-time collaboration, Power BI Service allows multiple users to work simultaneously	Awkward, requires cloud save or manual sharing
Connectivity	Multiple data sources (Excel, SQL, Google Analytics, Big Data, etc.)	Limited connectivity compared to Power BI
Data Handling	Can handle 100 million rows of data	Limited to 1 million rows of data
Integration	Seamless Excel integration, Power BI Embedded for apps and websites	Basic integration with other Microsoft products
Data Updates	Automatic updates based on real user experiences	Manual updates
Templates	Many built-in and customizable visuals	Wide variety of templates, easy customization
Reporting	Ideal for KPIs, dashboards, and visual reporting	Great for tabular reports
Weaknesses		
Complex Tables	Difficult to handle multiple linked tables	Handles complex, personalized calculation tables
User Interface	Busy UI with icons and options	Simpler UI, but lacks advanced visual elements
Visual Configuration	Limited customization of visuals	Visuals are customizable, but options are fewer
Learning Curve	DAX formula can be rigid and complex	Easier to learn formulas and functions
Complexity	Intuitive, but additional tools can increase complexity	Simpler for general tasks, more straightforward
Source Code	Closed-source (Microsoft Office suite)	Closed-source (Microsoft Office suite)

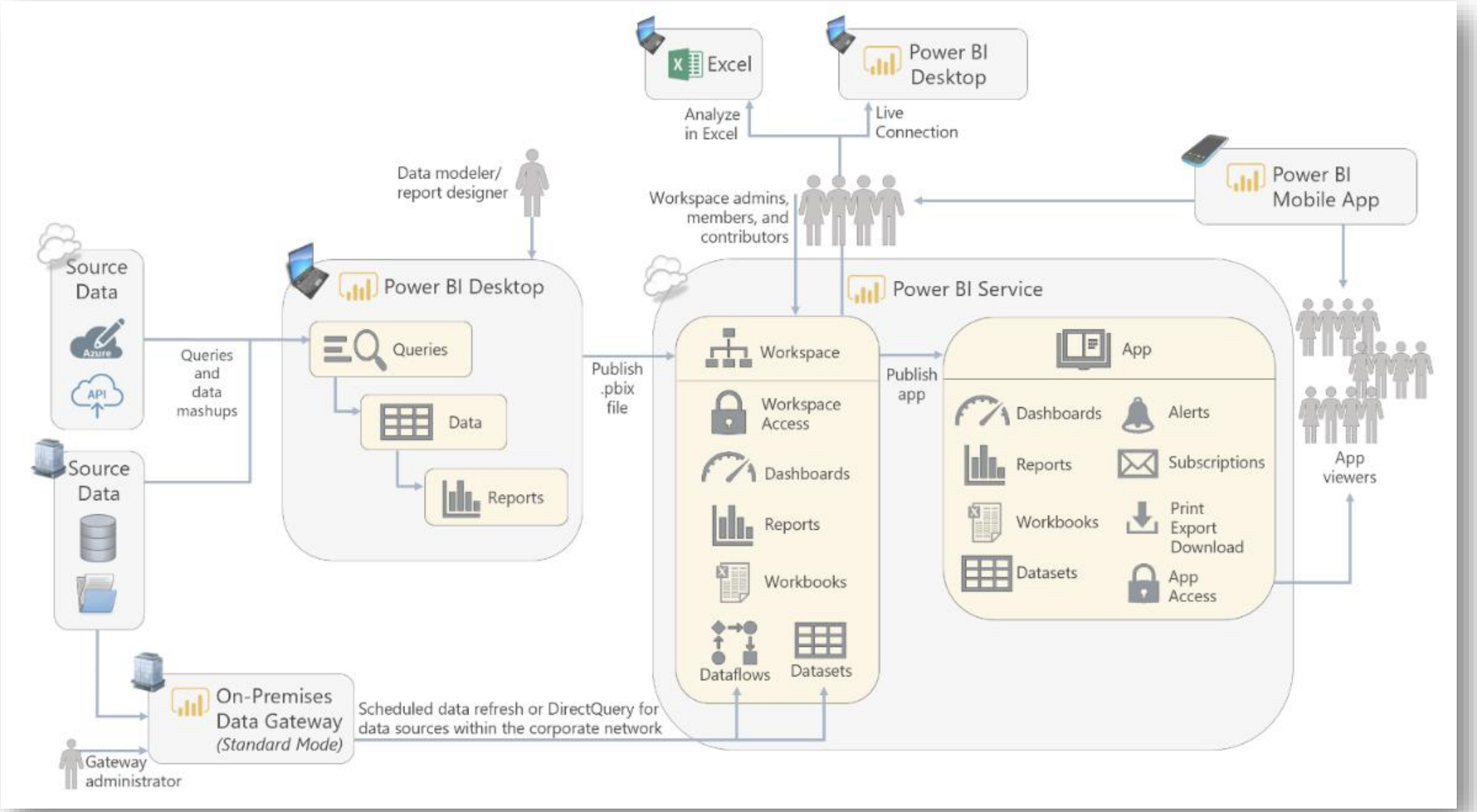
How Does Power BI Work?

- **Connecting to Data Sources:** It all starts with Power BI connecting with different kinds of data sources. Such may include databases, cloud services, Excel files or web APIs. Users can link several data sources at once so as to bring together all the organisation's data in one place.
- **Data Transformation and Importation:** As soon as you are connected, data is brought into Power BI. This data may need cleaning or transformation before it can be used. Through the Power Query tool in Power BI, users can filter, clean and reshape the data to make it suitable for analysis. Such a process may entail de-duplication, missing values handling or changing of data types.
- **Data Modeling:** After cleaning the data, it is loaded into the Power BI's data model. It entails setting up relationships between various tables of data. On the other hand, calculated columns, measures, and hierarchies can be made that help in summarising and analysing this information.
- **Visualisations Creation:** Once you have designed your Data Model on Power BI, for instance, you may now create visualisations by using charts which have many options such as maps, graphs, etc. to represent your information.
- **Reports & Dashboards Building:** Multiple visualisations are combined to create reports, which can span several pages. Dashboards are created by pinning key visualisations from these reports onto a single page, providing an overview of the most important metrics. These reports and dashboards are then published to the Power BI Service.
- **Collaborating and Acting on Insights**: Finally, the insights gained from Power BI reports and dashboards can be used to drive business decisions. They can discuss insights, comment on reports and make data-driven decisions as a team. Thus, Power BI enables people to explore their data for a better understanding of it in terms of trends or opportunities.

Power BI Data Flow



Power BI High level Data Flow



Power BI – Elements

1. Power Query: Power Query is the first step in working with data in Power BI. It allows you to connect to different data sources like Excel files, databases, web pages or even cloud-based services. Once the data is imported Power Query helps you to clean, transform and shape it into a useful format. We can remove unnecessary rows, change column names, combine data from multiple tables and much more all without programming using Power BI. This process is called “data wrangling" and Power Query makes it easy and visual.

2. Power View: Power View helps us to create visuals and reports by simply dragging and dropping data fields. It support wide range of chart types like bar charts, pie charts, line charts, maps and more. It’s interactive which means you can click on parts of the chart to filter other parts of the report. It is mainly used to explore your data visually and discover trends. It is helpful for users who prefer to see patterns through visuals rather than numbers.

3. Power BI Desktop: Power BI Desktop is the main platform where most of the work happens. It combines all the features like Power Query (for cleaning data), Power Pivot (for modeling data) and Power View (for visualizing data) into one tool. This is where you build complete reports and dashboards. This interface is user-friendly and it allows you to design everything offline before publishing it online. It is like workshop were we prepare everything here before sharing it with our team or organization.

4. Power BI Service: After we create our reports in Power BI Desktop you can publish them to the Power BI Service which is also known as the Power BI Website. Here we can share your reports with others, collaborate in teams and access your data from any device using a browser. It’s a cloud-based platform that also lets us schedule data refreshes and receive automatic updates. We can also create dashboards here by pinning visuals from different reports into one view for easy monitoring.

5. Power Pivot: Power Pivot is the engine behind the scenes that helps you work with large datasets quickly. It lets you create data models which means you can link different tables together and use formulas (called DAX formulas) to perform calculations. Even if you have millions of rows it compresses the data and processes it efficiently. This is great for building powerful reports without slowing down your system. It’s especially useful for users who want to do in-depth analysis.

6. Power Map: Power Map is used when your data includes geographical locations like countries, cities or coordinates. It creates 3D maps that let you view your data based on location. You can see patterns like sales by region, customer distribution or delivery routes. The maps are animated and you can even show changes over time. For example, how sales grew across different areas month by month.

7. Power Q&A: Power Q&A is like talking to your data using natural language. You don’t need to know any query language just type a question like “Show total sales for 2024” or “Top 5 selling products” and Power BI will understand and give you the answer instantly in the form of a chart or graph. It uses smart algorithms to understand your words and give meaningful answers. This feature makes data analysis more accessible to people who may not be experts.

8. Power BI Mobile Apps: Power BI has mobile apps for iPhone, Android and Windows devices. These apps allow you to access your reports and dashboards anywhere and anytime. You can view the latest data, check KPIs and stay updated on business performance while traveling or during meetings. The mobile interface is designed to be clean and responsive so you can scroll through your visuals easily even on a small screen.

Power BI – Installation

1. Open Microsoft Store and Search Power BI
2. Click on GET on download and install Power BI

Power BI – Dashboard

DAX in Power BI

What is DAX?

DAX or Data Analysis Expressions drive all the calculations you can perform in Power BI. DAX formulas are versatile, dynamic, and very powerful – they allow you to create new fields and tables in your model. DAX is a formula language used in Power BI, Power Pivot, and SSAS Tabular models

DAX formulas are made up of 3 core components, and this tutorial will cover each of these:

Syntax – Proper DAX syntax is made up of a variety of elements, some of which are common to all formulas.

Functions – DAX functions are predefined formulas that take some parameters and perform a specific calculation.

Context – DAX uses context to determine which rows should be used to perform a calculation.

Where are DAX Formulas Used in Power BI?

There are three ways you can use DAX formulas in Power BI:

Calculated Tables - These calculations will add an additional table to the report based on a formula.

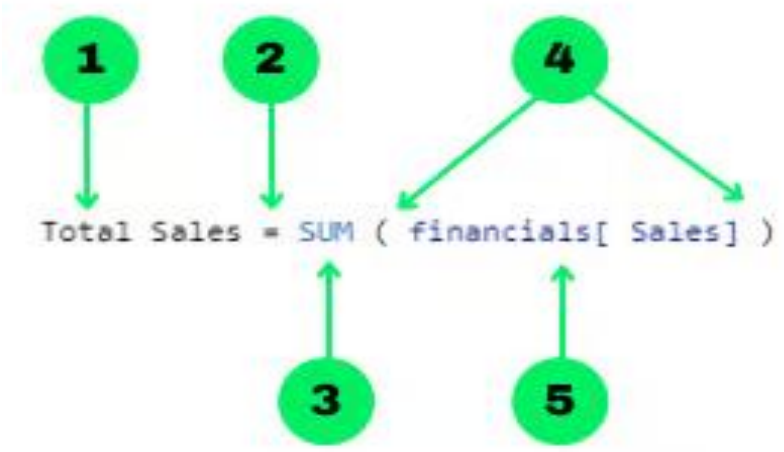
Calculated Columns - These calculations will add an additional column to a table based on a formula. These columns are treated like any other field in the table.

Measures - These calculations will add a summary or aggregated measure to a table based on a formula.

DAX in Power BI

How to Write a DAX Formula

1. The name of the measure or calculated column
2. The equal-to operator ("=") indicates the start of the formula
3. A DAX function
4. Opening (and closing) parentheses ("()")
5. Column and/or table references
6. Note that each subsequent parameter in a function is separated by a comma (",")



Some of the most common DAX functions used in reports are:

Simple calculations: COUNT, DISTINCTCOUNT, SUM, AVERAGE, MIN, MAX.

SUMMARISE: Returns a table typically used to further apply aggregations over different groupings.
CALCULATE: Performs an aggregation along with one or more filters. When you specify more than one filter, the function will perform the calculation where all filters are true.
IF: Based on a logical condition, it will return a different value if it is true or false. This is similar to the CASE WHEN operation in SQL.
IFERROR: Looks for any errors for an inner function and returns a specified result
ISBLANK: This function checks if the rows in a column are blank and returns true or false. It is useful in conjunction with other functions like IF.
EOMONTH: Returns the last day of the month of a given date (column reference in a date format) for as many months in the past or the future.
DATEDIFF: returns the difference between two dates (both as column references in date formats) in days, months, quarters, years, etc.

Power BI- Advanced Dashboard

Gauge

Table

Decomposition Tree

Q&A

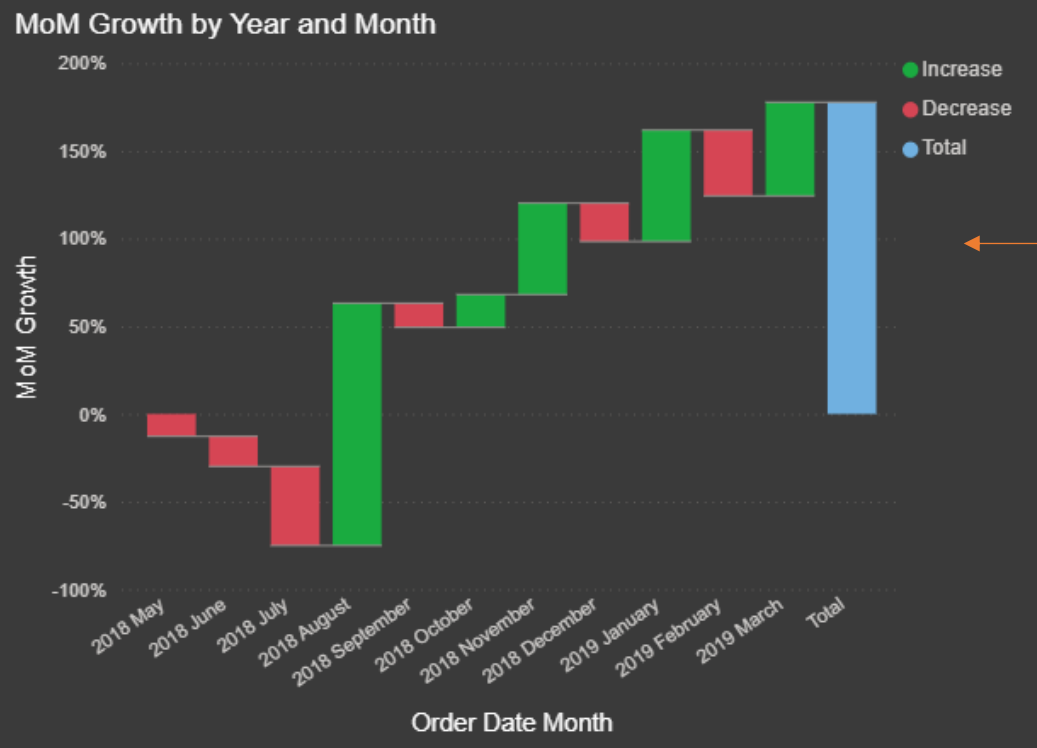
Treemap

Matrix

Power BI- Sales Analysis Dashboard

Table

Year	Month	Sum of Amount	Prev Month Amount	MoM Growth
2018	April	32726		
2018	May	28545	32726	-12.78%
2018	June	23658	28545	-17.12%
2018	July	12966	23658	-45.19%
2018	August	30899	12966	138.31%
2018	September	26628	30899	-13.82%
2018	October	31615	26628	18.73%
2018	November	48086	31615	52.10%
2018	December	37579	48086	-21.85%
2019	January	61439	37579	63.49%
2019	February	38424	61439	-37.46%
2019	March	58937	38424	53.39%
Total		**431502**	**372565**	**15.82%**



Waterfall chart

Power BI- Image and Rating Dashboard

Table

name
DHYANI Powerful Portable & High Power 12V Vacuum Cleaner for Car and Home Wet and Dry Car Vaccum Cleaner Multipurpose Vacc...
Digital Craft 2Pc 5 Inch 125mm Sponge Waxing Buffing Polishing Pad for Car Polisher Waxing
Digital Craft 7Inch Car Wave Sponge Polishing Pad Set Polishing Buffer Waxing Buffing Pad Polishing Sponge Wheel Vehicle P...
Digital Craft 8Mtr High Pressure Washer Accessories Hose Cleaning Extension Flexible Pipe with Gun, 3000 PSI Tools
Digital Craft Adjustable Speed Car Buffer Polishing Waxing Machine Furniture Ceramic Vehicle Surface Paint Care Washing Po...
DIKANSHA® Car Cleaning Duster, Ultra Soft Microfiber Brush Multipurpose Smooth Cleaner for Car Office Home Use with Storag...
DirtBurn 5 Pcs Car Cleaning Brush Kit Different Size Detailing 5 Different Sizes Plastic Handle Car Cleaning Accessories B...
DirtBurn Yellow Foam Wax Applicator Sponge Pad for Car Waxing Polish Detailing Care (Pack of 12pcs)
DIVINE Car Windshield Glass Concentrated Washer Tablets Solid Car Effervescent Tablets Glass Solid Wiper Cleaning Tablets ...
DIY Crafts 12-17x Gross Polishing Buffing Pad Kit Polisher M10/14 Universal Drill Adapter Car Sponge Wheel Kit Car Wash Cl...
DIY Crafts 32 Pcs Deep Scratch Remover Car All Type Glass Headlight Windshield Polishing Shining Buffing Kit 3'' Wheel Cer...
DIY Crafts Buffing Polishing Wheels with 3 Colors Polishing Compounds Kit, Including Flannel Cotton Yellow Buffing Wheel (...
DIY Crafts Fur Felt Pad Only for Car & Motorbike Care High Gross Wax Polishing Buffing Pad Car Polishing Buffer Drill Adap...
DIY Crafts M14 Stainless Steel Extension Shaft Rod Threaded for Rotary Polisher,Car Polisher,Polishing Backing

Image

Ratings

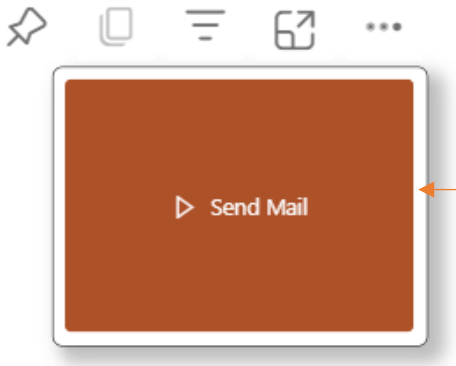
4.0

Rating

Power BI- Use of Power Automate

Table

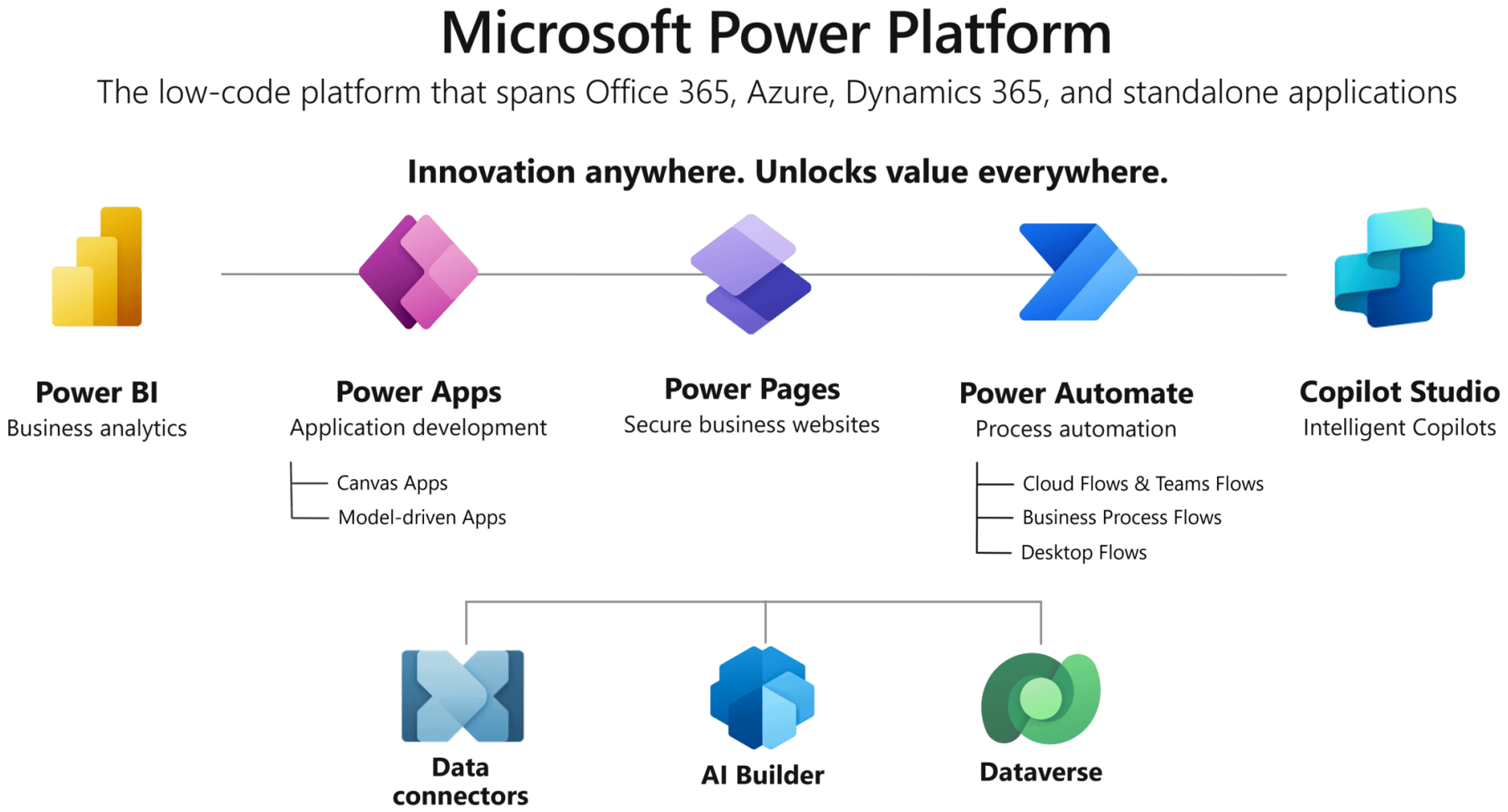
Customer Name	Customer Email	Ticket Priority	
Aaron Brown	tracyramirez@example.net	Low	Open
Aaron Carter	ycook@example.net	High	Open
Aaron Chavez	jjackson@example.com	Low	Pending Customer Response
Aaron Dixon	gary80@example.net	High	Pending Customer Response
Aaron Henry	marcmyers@example.com	Medium	Closed
Aaron Henry	willisjason@example.org	Low	Closed
Aaron Hensley	tamara40@example.org	Medium	Open
Aaron Kelly	tmitchell@example.net	Low	Closed
Aaron Martinez	youngcameron@example.org	High	Closed
Aaron Miller	perezdave@example.net	Medium	Pending Customer Response
Aaron Morales	jerry44@example.com	Medium	Open
Aaron Morales	tgregory@example.net	Critical	Closed
Aaron Nelson	jcrawford@example.net	High	Open
Aaron Norris	michaelbarbara@example.net	Medium	Pending Customer Response
Aaron Norris	scott33@example.org	High	Pending Customer Response
Aaron Petty	brandoncolon@example.org	Critical	Closed
Aaron Pope	nwilliams@example.net	Critical	Closed
Aaron Powell	jfitzgerald@example.com	Low	Closed
Aaron Reed	stacey08@example.org	High	Closed
Aaron Reese	dlindsey@example.org	Medium	Open
Aaron Roberts	vaughanbrenda@example.org	High	Pending Customer Response
Aaron Salazar	stephaniegarcia@example.net	Critical	Closed
Aaron Stephens	geoffrey94@example.net	Medium	Pending Customer Response
Abigail Davis	rodriguezchristine@example.net	Medium	Pending Customer Response
Abigail Harrison	diana50@example.org	Low	Closed
Abigail Jones	brentbradshaw@example.org	Medium	Pending Customer Response
Abigail Kennedy	millerjennifer@example.org	Low	Pending Customer Response



Power Automate

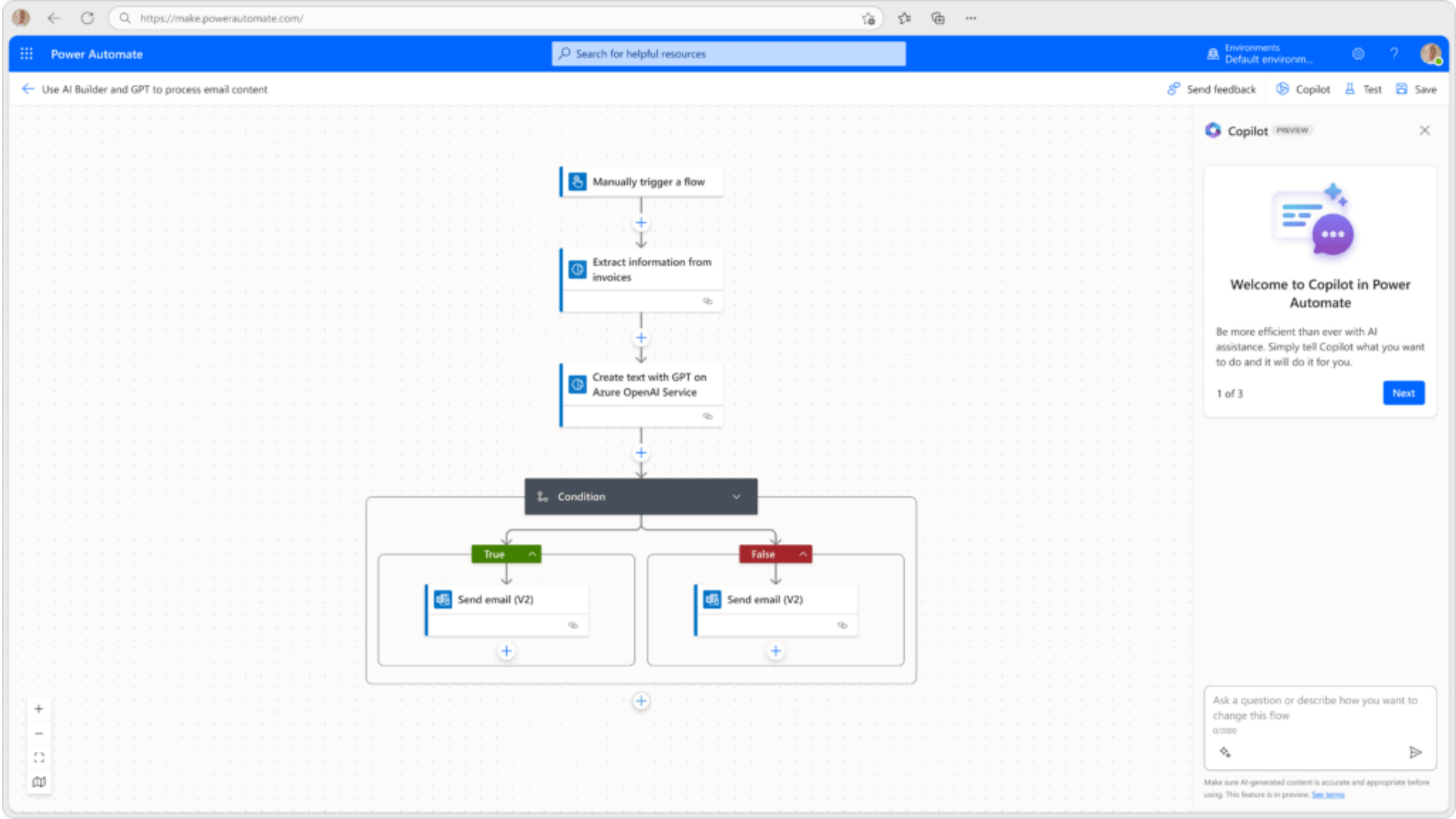
Power Platform

The Microsoft Power Platform is a suite, or an ecosystem, of low-code and no-code tools designed to help users build custom applications, automate workflows, analyze data, and create AI chat bots – all without needing extensive coding expertise.



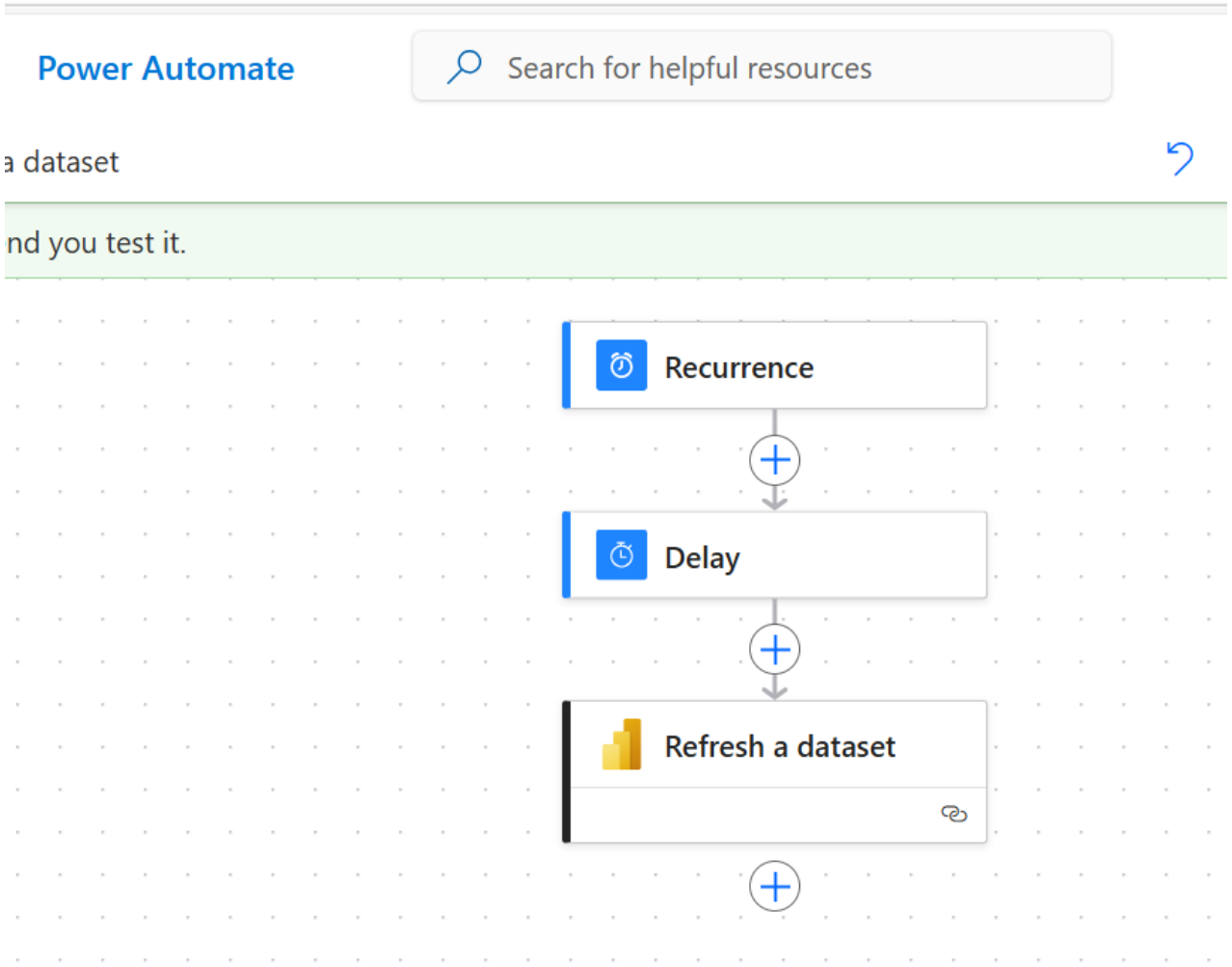
Power Automate

Power Automate is a Drag and Drop / low code solution that allows users to automate repetitive chores and business processes using feeds. It is built to create fast results and be easy to use. In use, it is possible to automate everything from small activities to large-scale systems, which results in the opportunity to have more time for more qualified work.



Cloud Flow Designer

Power BI Dataset Refresh by Power Automate



RLS

Row-level security (RLS) is a type of data governance that allows developers and admins to limit the data an end-user has visibility to within a Power BI report and/or dataset based on the logic applied at the row level.

Location Based RLS: An organization wants specific subsets of users only to see information about locations that are specific to them (City/State/Region/Country, etc.)

Ex: We are the VP of Finance for the Western region and should only see data for that specific area.

Employee Based RLS: A developer wants users only to see information that they are tied to or responsible for (Email/ Name)

Ex: We are a Business Development Representative and should only see data for CRM pipeline opportunities that fall under my name.

Business Line Based RLS: Users should only see information within a specific business line (Product/Service/Department).

Ex: We are a Buyer of men’s clothing and should only see data related to my business domain/division.

RLS

Example of DAX Query for the RLS in Country table (referring Day 5 recording)

```
VAR User=LOWER(USERPRINCIPALNAME())

var RLSValues=
SELECTCOLUMNS(FILTER(security_table1,security_table1[email]=User),"AllowedValues",security_table1[Country])

RETURN

countries[Country] in RLSValues
```

==========

Example of DAX Query for the RLS in Product table (referring Day 5 recording)

```
var User=LOWER(USERPRINCIPALNAME())

var RLSValues=
SELECTCOLUMNS(FILTER(security_table,security_table[email]=User),"AllowedValues",security_table[Product])

RETURN

product_table[Product] in RLSValues
```

==========

Example of some DAX queries

To get all cities from Customer Table (referring Day 5 recording)

```
EVALUATE
VALUES ( Customer[City] )
```

To get all rows which have the city “Redmond” from Customer Table (referring Day 5 recording)

```
EVALUATE
CALCULATETABLE ( Customer, Customer[City] = "Redmond" )
```

To get all rows which have the city start with letter “R” from Customer Table (referring Day 5 recording)

```
EVALUATE
CALCULATETABLE ( VALUES ( Customer[City] ), LEFT ( Customer[City], 1 ) = "R" )
```

To get all rows in ascending order which have the city start with letter “R” from Customer Table (referring Day 5 recording)

```
EVALUATE
CALCULATETABLE ( VALUES ( Customer[City] ), LEFT ( Customer[City], 1 ) = "R" )
ORDER BY Customer[City]
```

Conclusion

- We are at the End of the presentation. Hopefully we have learned some concepts about Power BI and it's usage in Practical Scenario.
- Although you can learn on Power BI apart from this presentation, so, I have motioned some tutorial sites for your reference.
- Tutorial Website:

https://www.geeksforgeeks.org/power-bi/power-bi-tutorial/

https://www.tutorialspoint.com/power_bi/power_bi_tutorial.pdf